Dong 2020 Prostate Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

The malignant manifold clusters BY SAMPLE. After Harmony(sample) the kNN graph is a single component and median iLISI rises 1.00→1.31 — the scalars a naive test reads as 'integrated'. But on the regenerated embedding each patient holds its own territory: self-neighbour enrichment 3.3×/7.3×/3.7×/1.7× (patients #1/#2/#5/#6); 3 of 4 cross the 2.5× segregation line, and patient #6 (small-cell NEPC, ~49% of cells) forms its own pole. Integration does not fix it — enrichment barely moves across θ 2→20 and iLISI plateaus at 1.31 (θ=50 collapses); holds across n_pcs 20/30/40. The geography is inter-sample clonal/CNV + histologic (adeno vs NEPC) heterogeneity, not a shared cell-state continuum.

Gate 2 — module geography → NA

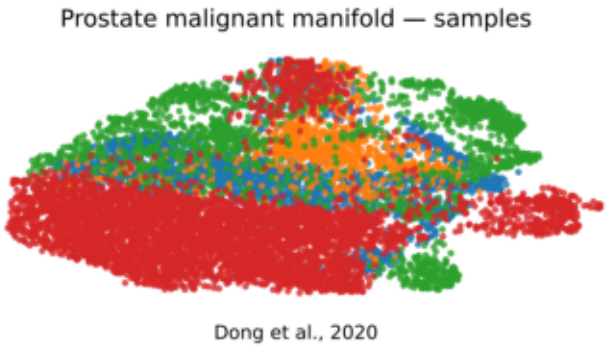
Not evaluated — Gate 1 NO-GO (chain stops). Every sample carries the same embryonic modules, but there is no shared sample-integrated manifold on which to fit a trajectory.

Gate 3 — hub cross-repression → NA

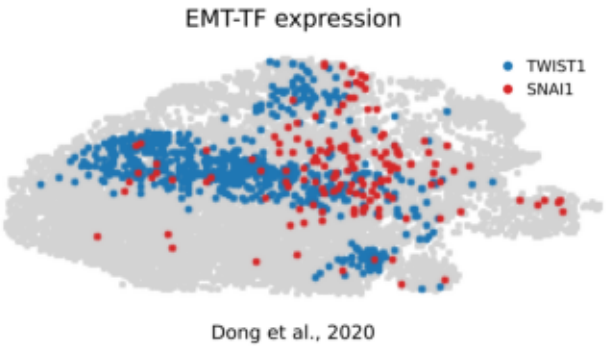
Not evaluated — requires a branching trajectory, which Gate 1 precludes.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Prostate adenocarcinoma + small-cell NEPC (Dong et al., 2020), 10x — 9,698 malignant cells (4 patients).

Verdict.
The embryonic-program pattern does NOT reproduce — the malignant compartment does not form a shared, sample-integrated cell-state manifold. It fragments into per-patient islands (clonal/CNV heterogeneity) plus a distinct small-cell NEPC pole (patient #6), so no trajectory is fit. Gate 1 NO-GO / Gates 2 & 3 NA. Descriptive EMT-TF overlay: TWIST1 (26.8%) · SNAI1 (6.0%).

Parameter choices

Min-cell threshold	floor 100 (all 4 samples kept, 915→4,765 cells); gap-finding rule REJECTED (premise unmet)	**Overdispersed genes**	1,670 (scFates GAM)
PCA / n_pcs	n_pcs=30 (Kürten 20 / Puram 40); Gate-1 call robust at 20/30/40	**Integration**	Harmony(sample); θ swept 2/5/10/20(+50); θ=10 best (max iLISI); none integrates
Geometry	None — no trajectory fit (Gate 1 NO-GO); per-sample archipelago	**EMT-TFs**	TWIST1 (26.8%) · SNAI1 (6.0%); 10-TF panel widened to FOXC/PRRX; VIM excluded